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Fish 460

Tripping on Aesthetascs paper

Abstract

Introduction

The intertidal zone facilitates extreme shifts in factors like temperature and salinity on a regular basis. Organisms that live within the intertidal, such as crabs have spent millennia adapting to the harsh, but predictable changes. Some have a thick carapace to retain water and prevent desiccation (CITATION). While others are competent diggers to stay in a moist environments (CITATION). While crabs, conform to outside temperatures because they're ectothermic, they still have to stay within an optimal range of temperatures they can withstand. The higher the temperature, the higher the mortality rate for most crab species (CIATATION).

The yellow shore crab, *Hemigrapsus oregonensis* is commonly found in the intertidal across the US west coast where it experiences regular temperature shifts. They are most used to cooler ocean temperatures, so their metabolisms are likely most used to cooler temperatures. As temperatures are projected to gradually increase, their metabolisms should likely increase as well. However, along with any other organisms, enzymes important for bodily processes can only survive within a specific window of heat, denaturing if it gets too hot. While some likely have different scopes compared to others, one part of *H. oregonensis* lifestyle that is necessary for their survival is scent. We seek to test if temperature affects the olfactory scope of *H. oregonensis*, with temperature likely having some effect on olfactory acuity. We also seek to test how the metabolism of *H. oregonensis* changes in relation to prolonged elevated temperatures, with elevated temperatures likely promoting faster metabolisms.

Methods

In this experiment, two groups of *H. oregonensis* were made, 6 in a control temperature of 18 Celsius, and 6 others in elevated temperatures of 24 C. Crabs were fed every other day, and fasted 2 days before experiments. Upon maze experimentation, each *H. oregonensis* was placed into a maze that consisted of two barriers that blocked the line of sight. They were given 1 minute to acclimate to their surroundings, then chunks of herring were placed on the opposite side of the maze from the crab. The time to find food was recorded for each crab. For predator response experiments, *H. oregonensis* were placed in a tank with a divider in the middle and given 1 minute to acclimate. After acclimation, *C. maenas* was placed on the side opposite of *H. oregonensis*, which were filmed for a minute to evaluate the amount and rate of antennule flicking.

For experiments directly measuring the metabolism, all *H. oregonensis* were placed into small containers filled with resazurin. Aliquots of resazurin was taken every half hour, with a total of two samples per crab. Resazurin was sampled all 3 days of experimentation. Blood hemolymph was extracted on the last day of experimentation, where it was extracted from each crab, and processed for interpretation.

Figures

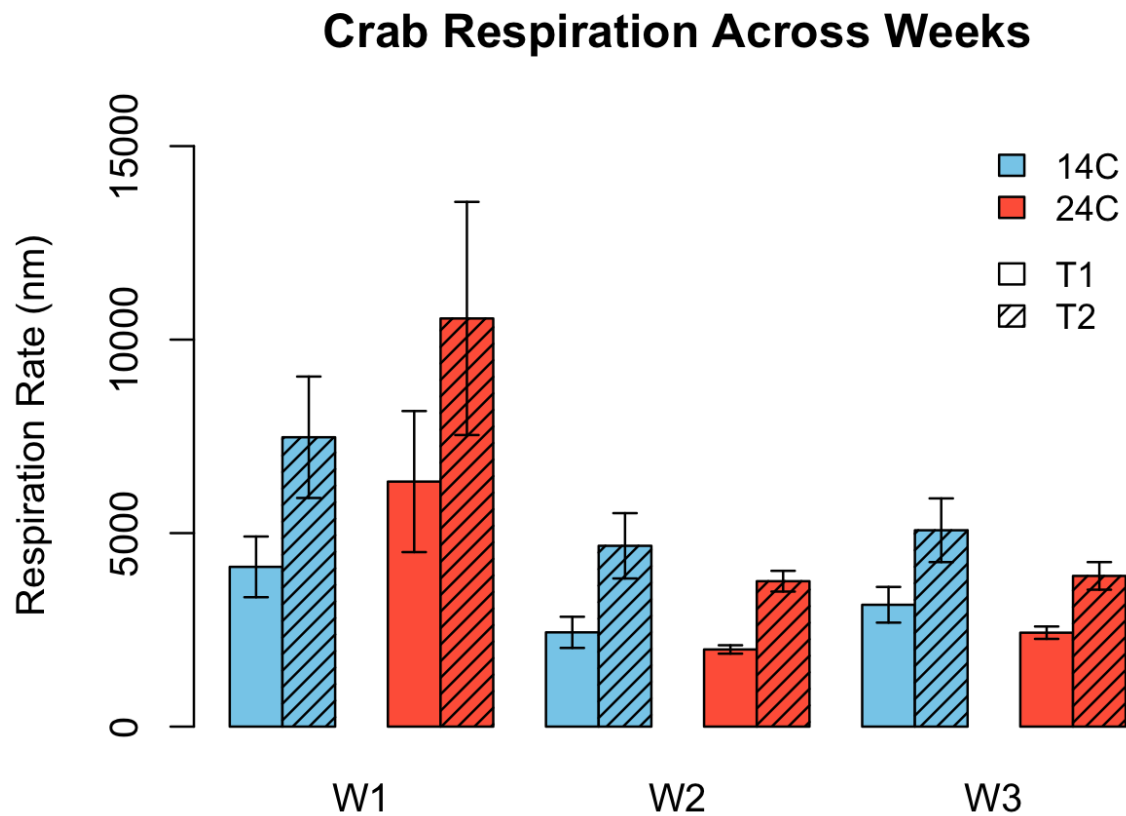


Figure 1. Respiration rate of *H. oregonensis* across testing periods. T1 is the first reading taken, and T2 is the second reading in the same week. Crab 12 was only part of week 1 in the 24C group, since it died between week 1 and week 2. It also had an extremely high respiration rate (reflected in the error bar), which may corroborate with it dying soon after, but that may have skewed the data for the 24C group in week 1.

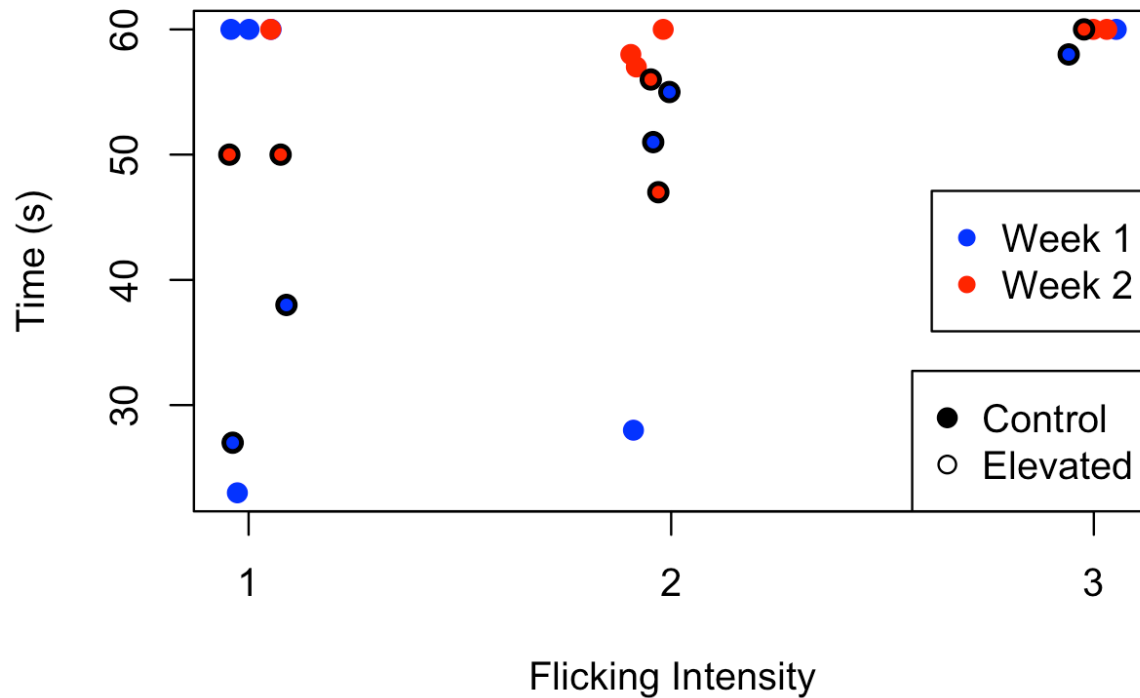


Figure 2. The time spent and intensity of antennule flicks across both treatments of crabs across both weeks.

Results

One crab in the 24C treatment died before maze and predator trials begun. Out of 22 trials, with 2 per *H. oregonensis* across weeks, food was found 86% of the time in the maze (FIG NEEDED). On average, it took each crab 94 seconds to find the food in the maze. For the predator experiment, the average time for antennule flicking was 51 seconds (FIG NEEDED). Forty percent flicked at a low intensity of 1, 36% with medium intensity of 2, and 22% at a fast intensity. For the metabolic experiments, the average rates of oxygen consumption for T1 and T2 were 2 times higher the first week than the other 2 weeks.

Discussion

The sample size prevented statistical significance with any of the experiments. Regardless, most crabs were able to locate the food when they weren't able to visually see it in the maze. Some crabs started scuttling around as soon as they were placed in the maze, whereas others didn't move until food was placed. While difficult to understand the scope of their olfactory acuity from the maze alone, personality differences between crabs is present based off how they reacted to being in the maze. For the predator experiment, no difference in antennule flicking rates was apparent, across the elevated and control temperature groups. Most crabs flicked their antennules the entire time, albeit some at faster rates than others. The crabs that were the most energetic tended to flick them the most, whereas the more lethargic crabs usually flicked them less. This neither supports nor denies our hypothesis. More data is needed to create a better understanding of how temperature plays with olfactory acuity.

For the metabolic experiments, the high metabolic rate for all the crabs at the beginning compared to the later data collection days may be due to the crabs being unfamiliar with their surroundings since they were collected only a couple days before. Though the sample size is small, in weeks 2 and 3, crabs in the control environments had slightly higher metabolic rates compared to the elevated treatment group. This goes against our hypothesis that the metabolism of *H. oregonensis* will increase with increased temperature. The crabs may be more adapted to living in warmer shallow waters than cooler waters to the point that their bodies have to work harder for their physiological processes to work adequately.